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Misperception of body weight and associated socioeconomic and health-related factors among Korean female adults: A nationwide population-based study

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Background: Misperception of body weight is associated with various psychological and health problems, including obesity, eating disorders, and mental problems. To date, female-specific risk factors, including socioeconomic or health-related lifestyle features, or their indicative performance for the misperception in Asian women according to age groups remain unknown.

Objectives: To investigate the prevalence and associated risk factors for the mismatch in self-perceived body weight and evaluated the classification performance of the identified risk factors across age groups in female adults.

Methods: We analyzed data of 22,121 women (age 19–97 years) from the 7-year Korea National Health and Nutrition Examination Survey dataset (2010-2016). We evaluated self-perceived body weight of the participants with their actual weight using the body mass index cut-off and grouped them by age: early adulthood (19–45), middle adulthood (46–59), and late adulthood (≥60). Logistic regression was conducted in each age group based on their weight misperception. The classification performance of the identified risk factors was evaluated with a bagging tree ensemble model with 5-fold cross-validation.

Results: 22.2% (n=4,916) of the study participants incorrectly perceived their body weight, of which 14.1% (n=3,110) and 8.2% (n=1,806) were in the underestimated and overestimated groups. Among the age groups, the proportion of participants who misperceived their body weight was highest in late adulthood (31.8%) and the rate of overestimation was highest in early adulthood (14.1%). We found that a lower education level, absence of menopause, perception of themselves as unhealthy, and efforts for weight management were significantly associated with the overall misperception (overestimation or underestimation) of body weight across age groups.

Based on the identified risk factors, the highest area under the receiver operating curve (AUROC) and accuracy of the best classification model (weight overestimation in all participants) were 0.758 and 0.761, respectively. Adding various associated lifestyle factors to the baseline model resulted in an average increase of 0.159 and 0.135 in AUROC for classifying weight underestimation and overestimation, respectively.

Conclusions: Age, education level, marital status, absence of menopause, amount of exercise, efforts for weight management (gain, loss, and maintenance), and self-perceived health status were significantly associated with the mismatch of body weight.

KEYWORDS

perceived weight, weight misperception, weight management, health informatics, body mass index, public health

1 Introduction

Obesity is a complex and multifactorial disease that is a major public health issue. The prevalence of obesity has been epidemically increasing worldwide; being overweight and obese affects over one-third of the world's population (1), and it is expected that 38% of the world's adult population will be overweight, and 20% will be obese by 2030 (2). Excess weight and obesity are well-established critical risk factors for comorbidities of chronic diseases such as type 2 diabetes, hypertension, dyslipidemia, cardiovascular events, or premature mortality (3, 4). Obesity represents a challenge in public health given its high prevalence, increasing trend, and accompanying health risks (2).

Issues of obesity and its related health problems have generated an increased attention to the perception of and satisfaction with body image and shape. Body image distortion can lead to inappropriate diet restriction, eating disorders, and excessive weight loss, which in turn, give rise to reproductive function issues, such as menstrual irregularity, hypothalamic dysfunction, infertility, and bone loss due to impaired sex hormone metabolism (5–9). Moreover, body dissatisfaction is linked to adverse psychological consequences such as poor self-esteem, depression, and anxiety (10–12).

Existing studies report certain factors associated with the discrepancy between perceived and actual body shape and size (13–15). Significant gender and age differences have been reported for this discrepancy; males and younger individuals tend to underestimate their weight (and vice versa for females) (14–17). Studies have explored comprehensive lifestyle factors, ranging from health-related behaviors to socioeconomic status; however, most of this research used cross-sectional designs, highly selected cohorts, or populations with specific age groups (e.g., high school or university students, and young adults) (18–22). In addition, there is growing attention on how to identify the target populations with weight misperception, and only a few studies have performed predictive analytics based on the associated behavioral and social factors that lead to body weight misperception.

This study investigated the perceived body weight and/or its disturbance and trends across age groups in Korean women, using large population data from the national registry to determine the associated female-specific and age-specific risk factors for the discrepancy between actual body size and self-perceived body weight. One of the main purposes of the present study was to predict the future occurrence of body weight misperception with a machine-learning approach, by providing data that could be used by the public health branch management for preventive interventions.

2 Materials and methods

2.1 Study population

Data from 22,121 Korean women aged 19–97 years (mean age 50.4 ± 16.6) included in this study were collected from the Korea National Health and Nutrition Examination Survey (KNHANES) performed between 2010 and 2016. The KNHANES is a nationwide survey that has been conducted by the Division of Chronic Disease Surveillance under the Korea Centers for Disease Control and Prevention since 1998. This annual program evaluates health-related behaviors and nutritional status assessments for representative South Korean populations. Specialized research teams that consist of selected professional investigators, including nurses, nutritionists, and